# Synthetic Difference in Differences

## Critical paper talk (group meeting)

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arXiv:1812.09970 (July 2021 version)

Presenter / 2026-10-17

# The aim is causal: ATT for a block-treated panel

$$\tau = \frac{1}{N_{\text{tr}}\, T_{\text{post}}} \sum_{i \in \text{tr}} \sum_{t \in \text{post}} \big( Y_{it}(1) - Y_{it}(0) \big)$$

- Unit, time, intervention: states, years, a single adoption date.
- Presenter's reading of Eq. 4.2 (block assignment). Staggered adoption is backup, not this talk.

# Paper card (presenter's filling)

| Field | This paper |
|---|---|
| Estimand | Block ATT (Eq. 4.2); labelled if vague |
| Identification | Low-rank $L$ plus $E[E \mid W, L] = 0$ |
| Key assumption | Assignment may depend on $L$, not on $E$ |
| Main result | CA: SDID $-15.6$ (se $8.4$), Table 1 |
| Biggest threat | Placebo designs are zero-effect |

# Identification: knowledge vs convenience

- A1 outcome $Y = L + W \circ \tau + E$, $L$ low rank (knowledge).
- A2 $E[E \mid W, L] = 0$ (knowledge; untestable).
- A3 unit-independence of $E$, homoskedasticity (convenience for placebo se).

$$E[E \mid W, L] = 0$$

## What breaks

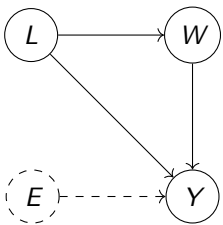
If assignment depends on $E$, $\tau$ is not identified; a better estimator does not close that gap.

# Method: unit weights, time weights, then TWFE

$$(\hat{\tau}^{\text{sdid}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min \sum_{i,t} \omega_i \lambda_t \big(Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau\big)^2$$

- $\omega$: pre-treatment trends parallel, intercept allowed.
- $\lambda$: pre periods that predict the post mean.

# DAG sketch: assignment may use $L$, not $E$



# California pack-sales estimate (Table 1)

| Estimator | $\hat{\tau}$ | se (placebo) |
|---|---|---|
| SDID | $-15.6$ | $8.4$ |
| SC | $-19.6$ | $9.9$ |
| DID | $-27.3$ | $17.7$ |

- Units: packs per person per year, 1970–2000, 38 controls.

# Causal reading holds only under A1–A2

- Under A1–A2, SDID estimates the block ATT, not DID's ATT under parallel trends.
- Sensitivity: none reported for assignment-on-*E*.
- Placebo RMSE (CPS baseline): SDID $0.028$, DID $0.049$ (Table 2, 1000 draws). Zero-effect design.

$$\widehat{\mathrm{RMSE}}_{\mathrm{CPS}}(\mathrm{SDID}) = 0.028$$

# Link to our project (fill or TODO)

- If we have few treated units and doubtful parallel trends, SDID is a robustness check beside DID and SC.
- Confirm a common adoption date before using Eq. 4.2.
- Project RQ: TODO.

# Reproducibility: violate A2 in a placebo

$$W_i = \mathbf{1}\{E_{i1} > \text{median}(E_{\cdot 1})\}$$

- Simulate assignment that depends on $E$; report bias of SDID vs DID vs SC (TODO: not run in this skeleton).
- Optional 60-min segment: replicate Table 1 side by side.

# What would change my mind

- A non-zero-effect simulation where SDID bias exceeds DID.
- Formal conditions for staggered adoption (appendix §8 is a recipe, not a theorem in the quickread).

$$\hat{\tau}^{\mathrm{sdid}} - \hat{\tau}^{\mathrm{did}} \quad \text{is not a test of } E[E \mid W, L] = 0.$$

**Limit**

Theory in the talk is block assignment only.

# Discussion: should we treat SC/SDID mismatch as a warning?

- Paper: when SC and SDID disagree, SDID is usually more credible (§2.2). Quickread tag: conjectured.
- California has no ground truth, so a mismatch alone does not tell us which estimator is wrong.

# References

| Source | Use |
|---|---|
| arXiv:1812.09970 | Tables 1–2 (quickread card) |
| arXiv:2305.18793 | Ding notation defaults |

# Thank you

$$\tau = E[Y(1) - Y(0) \mid \text{block assignment}]$$

# Backup: staggered adoption is not this estimand

$$\hat{\tau} = \sum_{g} \pi_g \hat{\tau}_g^{\text{sdid}}$$